Lesson 3 Teacher's Key

Godot Engine

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

Two scripts: one attached to bullet.tscn's Sprite2D root, one added to the Player's existing script:

```
# bullet.tscn's script:
extends Sprite2D
var speed := 400.0

func _process(delta):
    position.y -= speed * delta
    if position.y < -20:
        queue_free()

# Player's script - add near the top and inside _process:
var bullet_scene := preload("res://bullet.tscn")

func _process(delta):
    if Input.is_action_just_pressed("ui_select"):
        var b := bullet_scene.instantiate()
        b.global_position = global_position
        get_parent().add_child(b)
```

Level 2 - Complete Worked Solution

is_action_just_pressed() is already edge-triggered (true for exactly one frame per physical press), so Level 1's fire check needs no changes for Level 2. An optional tunable cooldown, if a slower fire rate is wanted:

```
var fire_cooldown := 0.0

func _process(delta):
    fire_cooldown -= delta
    if Input.is_action_just_pressed("ui_select") and fire_cooldown <= 0.0:
        var b := bullet_scene.instantiate()
        b.global_position = global_position
        get_parent().add_child(b)
        fire_cooldown = 0.2
```

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on fire

Add an AudioStreamPlayer2D node to the Player, Stream set to sfx_shoot.wav:

```
if Input.is_action_just_pressed("ui_select") and fire_cooldown <= 0.0:
    var b := bullet_scene.instantiate()
    b.global_position = global_position
    get_parent().add_child(b)
    fire_cooldown = 0.2
    $AudioStreamPlayer2D.play()
```

Visual: give the bullet its own look

Set modulate in the bullet's own _ready() so every spawned copy looks distinct without touching the Player's script at all - then count a short flash_timer back down in _process(delta) so the flash actually fades instead of staying permanent:

```
# bullet.tscn's script:
var flash_timer := 0.1

func _ready():
    modulate = Color.YELLOW
    scale = Vector2(1.3, 1.3)

func _process(delta):
    if flash_timer > 0.0:
        flash_timer -= delta
        if flash_timer <= 0.0:
            modulate = Color.WHITE
            scale = Vector2(1, 1)
```

Structure: keep bullet behavior in bullet.tscn's own script

The Player's script should only ever trigger a shot (instantiate, position, cooldown, sound) - everything about what a bullet actually does once it exists (movement, off-screen cleanup, its own look) belongs in bullet.tscn's own script. Keeping this boundary clean is what let the Sound and Visual ideas above each drop into one file without touching the other.

Complete Level 3 solution (all 3 goals combined)

Everything above combined, across the Player's script and bullet.tscn's own script:

```
# bullet.tscn's script:
extends Sprite2D
var speed := 400.0
var flash_timer := 0.1

func _ready():
    modulate = Color.YELLOW
    scale = Vector2(1.3, 1.3)

func _process(delta):
    position.y -= speed * delta
```

```
    if position.y < -20:
        queue_free()
    if flash_timer > 0.0:
        flash_timer -= delta
        if flash_timer <= 0.0:
            modulate = Color.WHITE
            scale = Vector2(1, 1)

# Player's script:
var bullet_scene := preload("res://bullet.tscn")
var fire_cooldown := 0.0

func _process(delta):
    fire_cooldown -= delta
    if Input.is_action_just_pressed("ui_select") and fire_cooldown <= 0.0:
        var b := bullet_scene.instantiate()
        b.global_position = global_position
        get_parent().add_child(b)
        fire_cooldown = 0.2
        $AudioStreamPlayer2D.play()
```